INCH-POUND

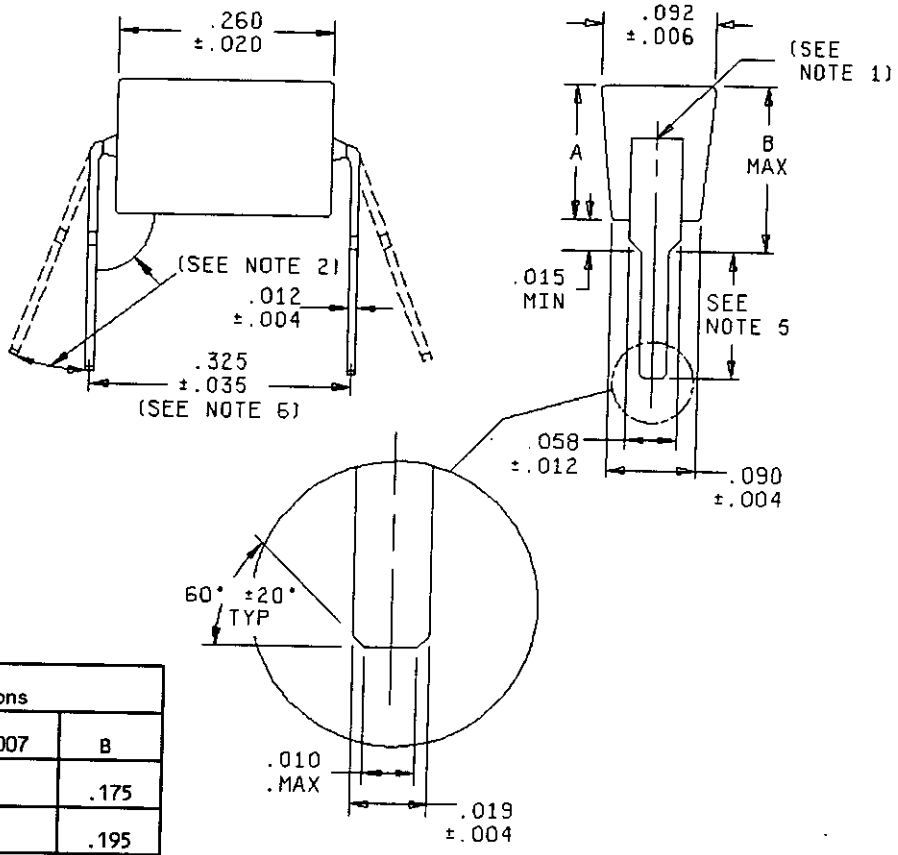
MIL-C-39014/22C
6 May 1993
SUPERSEDING
MIL-C-39014/22B
9 August 1982

MILITARY SPECIFICATION SHEET

# CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY, STYLES CKR22, CKR23, AND CKR24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39014.




Ⓒ

| Dimensions | | |
|---|---|---|
| Style | A ±.007 | B |
| CKR22 | .128 | .175 |
| CKR23 | .155 | .195 |
| CKR24 | .283 | .320 |

FIGURE 1. Style CKR22, CKR23, and CKR24 capacitors.

Ⓒ denotes changes



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| Inches | mm | Inches | mm |
|---|---|---|---|
| .004 | 0.10 | .092 | 2.34 |
| .006 | 0.15 | .128 | 3.25 |
| .010 | 0.25 | .155 | 3.94 |
| .012 | 0.30 | .165 | 4.19 |
| .015 | 0.38 | .185 | 4.70 |
| .019 | 0.48 | .260 | 6.60 |
| .020 | 0.51 | .283 | 7.19 |
| .035 | 0.89 | .320 | 8.13 |
| .058 | 1.47 | .325 | 8.26 |
| .090 | 2.29 | | |

NOTES:
1. Leads shall be centered within ±.005 inches (0.13 mm).
2. The angle shall be 95° +10° -5°.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
5. .140 ±.020 inch (3.56 ±0.51 mm) for standard lead length; .170 ±.010 inch (4.32 ±0.25 mm) for optional longer lead length.
6. The distance between the centers of the mounting holes will be .300 ±.010 inches (7.62 ±0.25 mm).
7. Nonconductive material shall not extend beyond .030 inches (0.76 mm) from the edge of the capacitor body.

FIGURE 1. Style CKR22, CKR23, and CKR24 capacitors - Continued.



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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Molded plastic encapsulated.

Failure rate level: M, P, R, and S.

Capacitance value: See table III.

Capacitance tolerance: D = ±.5 pF, F = ±1 percent, J = ±5 percent, K = ±10 percent, and M = ±20 percent.

Operating temperature range: -55° to +125°C.

Voltage rating: See table III.

Dissipation factor: See table I.

Dielectric withstanding voltage: Body insulation shall be able to withstand 500 V dc between leads and case.

Insulation resistance:

At 25°C: Not less than 100,000 megohms or 1,000 megohm-microfarads, whichever is less.
At 125°C: Not less than 10,000 megohms or 100 megohm-microfarads, whichever is less.

Insertion loss: Not applicable.

Voltage-temperature limits: See table I. For capacitors with values of 18 pF or less, the voltage-temperature limits tolerance shall be determined in accordance with table II.

TABLE I. Dissipation factors and voltage-temperature limits.

| Capacitance change with reference to 25°C | | | |
|---|---|---|---|
| Characteristic symbol | Steps A through D table V of MIL-C-39014 Bias = 0 volts | Steps E through G table V of MIL-C-39014 Bias = rated voltage | Dissipation factor (percent) |
| CG<br>CH | 0 ±30 ppm/°C<br>0 ±60 ppm/°C 1/ | 0 ±30 ppm/°C<br>0 ±60 ppm/°C 1/ | .15<br>.15 |
| BX | ±15 percent | +15, -25 percent as applicable | 2.5 |
| BR | ±15 percent | +15, -40 percent as applicable | 3.0 |

1/ See table II for capacitance values less than 20 pF.



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TABLE II. Voltage-temperature limits for capacitance values of 18 pF or Less.

| Voltage-Temperature Limits for Capacitors Values of 18pF or less | |
|---|---|
| Capacitance Values (pF) | Voltage-Temperature Coefficient Tolerance |
| 1.0 - 1.8 | Not Measureable |
| 2.2 - 3.9 | 0 ± 250 PPM/°C |
| 4.7 - 6.8 | 0 ± 120 PPM/°C |
| 8.2 - 18.0 | 0 ± 60 PPM/°C |

Immersion:

IR - Not less than 50 percent of initial 25°C requirements.

Resistance to soldering heat:

ΔCap: ±5 percent or .25 pF, whichever is greater for CG and CH characteristics.
ΔC +15, -5 percent of the initial 25°C measurement for characteristics BX and BR.

DF: Not to exceed the initial limit.

IR: Not less than initial 25°C limit.

Terminal strength: MIL-STD-202, method 211, condition B, except that each lead shall be bent away from the body 90° from the original position (see figure 1) and back, two bends.

Life:

Rated conditions, 100 percent of dc rated voltage at 125°C.

0 through 3,000 hours:

IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.

ΔC - ±5 percent or .5 pF, whichever is greater from initial measured value for CG and CH characteristics.
±20 percent from initial measured value for characteristics BX and BR.

Dissipation factor - Not to exceed the initial limit.

4,000 through 32,000 hours:

IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.

ΔC - ±5 percent or .5 pF, whichever is greater from initial measured value for CG and CH characteristics.
±20 percent from initial measured value for characteristics BX and BR.

Dissipation factor - Not to exceed the initial limit.



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Accelerated condition, 200 percent of dc rated voltage at 125°C.

0 and 250 hours:

IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.

ΔC - ±5 percent or .5 pF, whichever is greater from initial measured value for CG and CH characteristics.
±20 percent from initial measured value for characteristics BX and BR.

Dissipation factor - Not to exceed the initial limit.

1,000, 2,000, and 4,000 hours:

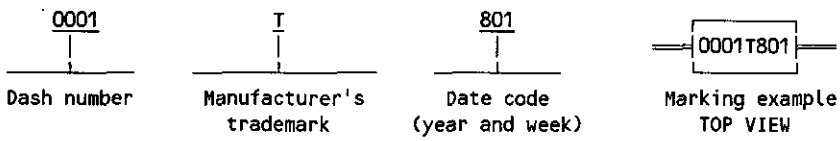
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.

ΔC - ±20 percent from initial measured value.

Dissipation factor - Not to exceed the initial limit.

Part or Identifying Number (PIN): M39014/22- (dash numbers from table III), i.e. M39014/22-0001.

Marking: In accordance with MIL-C-39014, except that minimum marking as shown in the following example, shall be used. Full marking shall be included on the package.






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© TABLE III. Dash numbers and electrical characteristics.

| Dash number Failure rate level (%/1,000 hours) | | | | | | | | Capacitance (pF) | Capacitance tolerance | DC rated voltage volts) |
|---|---|---|---|---|---|---|---|---|---|---|
| Standard lead length 1/ | | | | Optional longer lead length 1/ | | | | | | |
| 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | | | |
| Style CKR22, see table II for voltage-temperature limits. | | | | | | | | | | |
| 0001 | 0301 | 0601 | 0901 | 3001 | 3301 | 3601 | 3901 | 1.0 | D | 200 |
| 0004 | 0304 | 0604 | 0904 | 3004 | 3304 | 3604 | 3904 | 1.2 | D | " |
| 0007 | 0307 | 0607 | 0907 | 3007 | 3307 | 3607 | 3907 | 1.5 | D | " |
| 0010 | 0310 | 0610 | 0910 | 3010 | 3310 | 3610 | 3910 | 1.8 | D | " |
| 0013 | 0313 | 0613 | 0913 | 3013 | 3313 | 3613 | 3913 | 2.2 | D | " |
| 0016 | 0316 | 0616 | 0916 | 3016 | 3316 | 3616 | 3916 | 2.7 | D | " |
| 0019 | 0319 | 0619 | 0919 | 3019 | 3319 | 3619 | 3919 | 3.3 | D | " |
| 0022 | 0322 | 0622 | 0922 | 3022 | 3322 | 3622 | 3922 | 3.9 | D | " |
| 0025 | 0325 | 0625 | 0925 | 3025 | 3325 | 3625 | 3925 | 4.7 | D | " |
| 0028 | 0328 | 0628 | 0928 | 3028 | 3328 | 3628 | 3928 | 5.6 | D | " |
| 0031 | 0331 | 0631 | 0931 | 3031 | 3331 | 3631 | 3931 | 6.8 | D | " |
| 0034 | 0334 | 0634 | 0934 | 3034 | 3334 | 3634 | 3934 | 8.2 | D | " |
| 0037 | 0337 | 0637 | 0937 | 3037 | 3337 | 3637 | 3937 | 10 | D | " |
| 0038 | 0338 | 0638 | 0938 | 3038 | 3338 | 3638 | 3938 | 10 | J | " |
| 0039 | 0339 | 0639 | 0939 | 3039 | 3339 | 3639 | 3939 | 10 | K | " |
| 0040 | 0340 | 0640 | 0940 | 3040 | 3340 | 3640 | 3940 | 12 | D | " |
| 0041 | 0341 | 0641 | 0941 | 3041 | 3341 | 3641 | 3941 | 12 | J | " |
| 0042 | 0342 | 0642 | 0942 | 3042 | 3342 | 3642 | 3942 | 12 | K | " |
| 0043 | 0343 | 0643 | 0943 | 3043 | 3343 | 3643 | 3943 | 15 | D | " |
| 0044 | 0344 | 0644 | 0944 | 3044 | 3344 | 3644 | 3944 | 15 | J | " |
| 0045 | 0345 | 0645 | 0945 | 3045 | 3345 | 3645 | 3945 | 15 | K | " |
| 0046 | 0346 | 0646 | 0946 | 3046 | 3346 | 3646 | 3946 | 18 | D | " |
| 0047 | 0347 | 0647 | 0947 | 3047 | 3347 | 3647 | 3947 | 18 | J | " |
| 0048 | 0348 | 0648 | 0948 | 3048 | 3348 | 3648 | 3948 | 18 | K | " |
| Style CKR22, voltage-temperature limits of 0 ±30 ppm/°C. | | | | | | | | | | |
| 0049 | 0349 | 0649 | 0949 | 3049 | 3349 | 3649 | 3949 | 22 | D | 200 |
| 0050 | 0350 | 0650 | 0950 | 3050 | 3350 | 3650 | 3950 | 22 | J | " |
| 0051 | 0351 | 0651 | 0951 | 3051 | 3351 | 3651 | 3951 | 22 | K | " |
| 0052 | 0352 | 0652 | 0952 | 3052 | 3352 | 3652 | 3952 | 27 | D | " |
| 0053 | 0353 | 0653 | 0953 | 3053 | 3353 | 3653 | 3953 | 27 | J | " |
| 0054 | 0354 | 0654 | 0954 | 3054 | 3354 | 3654 | 3954 | 27 | K | " |
| 0055 | 0355 | 0655 | 0955 | 3055 | 3355 | 3655 | 3955 | 33 | D | " |
| 0056 | 0356 | 0656 | 0956 | 3056 | 3356 | 3656 | 3956 | 33 | J | " |
| 0057 | 0357 | 0657 | 0957 | 3057 | 3357 | 3657 | 3957 | 33 | K | " |
| 0058 | 0358 | 0658 | 0958 | 3058 | 3358 | 3658 | 3958 | 39 | D | " |
| 0059 | 0359 | 0659 | 0959 | 3059 | 3359 | 3659 | 3959 | 39 | J | " |
| 0060 | 0360 | 0660 | 0960 | 3060 | 3360 | 3660 | 3960 | 39 | K | " |
| 0061 | 0361 | 0661 | 0961 | 3061 | 3361 | 3661 | 3961 | 47 | D | " |
| 0062 | 0362 | 0662 | 0962 | 3062 | 3362 | 3662 | 3962 | 47 | J | " |
| 0063 | 0363 | 0663 | 0963 | 3063 | 3363 | 3663 | 3963 | 47 | K | " |
| 0064 | 0364 | 0664 | 0964 | 3064 | 3364 | 3664 | 3964 | 56 | D | " |
| 0065 | 0365 | 0665 | 0965 | 3065 | 3365 | 3665 | 3965 | 56 | J | " |
| 0066 | 0366 | 0666 | 0966 | 3066 | 3366 | 3666 | 3966 | 56 | K | " |
| 0067 | 0367 | 0667 | 0967 | 3067 | 3367 | 3667 | 3967 | 68 | F | " |
| 0068 | 0368 | 0668 | 0968 | 3068 | 3368 | 3668 | 3968 | 68 | J | " |
| 0069 | 0369 | 0669 | 0969 | 3069 | 3369 | 3669 | 3969 | 68 | K | " |
| 0070 | 0370 | 0670 | 0970 | 3070 | 3370 | 3670 | 3970 | 82 | F | " |
| 0071 | 0371 | 0671 | 0971 | 3071 | 3371 | 3671 | 3971 | 82 | J | " |
| 0072 | 0372 | 0672 | 0972 | 3072 | 3372 | 3672 | 3972 | 82 | K | " |
| 0073 | 0373 | 0673 | 0973 | 3073 | 3373 | 3673 | 3973 | 100 | F | " |
| 0074 | 0374 | 0674 | 0974 | 3074 | 3374 | 3674 | 3974 | 100 | J | " |
| 0075 | 0375 | 0675 | 0975 | 3075 | 3375 | 3675 | 3975 | 100 | K | " |
| 0076 | 0376 | 0676 | 0976 | 3076 | 3376 | 3676 | 3976 | 120 | F | " |
| 0077 | 0377 | 0677 | 0977 | 3077 | 3377 | 3677 | 3977 | 120 | J | " |
| 0078 | 0378 | 0678 | 0978 | 3078 | 3378 | 3678 | 3978 | 120 | K | " |
| 0079 | 0379 | 0679 | 0979 | 3079 | 3379 | 3679 | 3979 | 150 | F | " |

See footnote at end of table.



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(C) TABLE III. Dash numbers and electrical characteristics - Continued.

| Dash number Failure rate level (%/1,000 hours) | | | | | | | | Capacitance (pF) | Capacitance tolerance | DC rated voltage volts) |
|---|---|---|---|---|---|---|---|---|---|---|
| Standard lead length 1/ | | | | Optional longer lead length 1/ | | | | | | |
| 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | | | |
| Style CKR22, voltage-temperature limits of 0 ±30 ppm/°C - Continued. | | | | | | | | | | |
| 0080 | 0380 | 0680 | 0980 | 3080 | 3380 | 3680 | 3980 | 150 | J | 200 |
| 0081 | 0381 | 0681 | 0981 | 3081 | 3381 | 3681 | 3981 | 150 | K | " |
| 0082 | 0382 | 0682 | 0982 | 3082 | 3382 | 3682 | 3982 | 180 | F | " |
| 0083 | 0383 | 0683 | 0983 | 3083 | 3383 | 3683 | 3983 | 180 | J | " |
| 0084 | 0384 | 0684 | 0984 | 3084 | 3384 | 3684 | 3984 | 180 | K | " |
| 0085 | 0385 | 0685 | 0985 | 3085 | 3385 | 3685 | 3985 | 220 | F | " |
| 0086 | 0386 | 0686 | 0986 | 3086 | 3386 | 3686 | 3986 | 220 | J | " |
| 0087 | 0387 | 0687 | 0987 | 3087 | 3387 | 3687 | 3987 | 220 | K | " |
| 0088 | 0388 | 0688 | 0988 | 3088 | 3388 | 3688 | 3988 | 270 | F | " |
| 0089 | 0389 | 0689 | 0989 | 3089 | 3389 | 3689 | 3989 | 270 | J | " |
| 0090 | 0390 | 0690 | 0990 | 3090 | 3390 | 3690 | 3990 | 270 | K | " |
| 0091 | 0391 | 0691 | 0991 | 3091 | 3391 | 3691 | 3991 | 330 | F | " |
| 0092 | 0392 | 0692 | 0992 | 3092 | 3392 | 3692 | 3992 | 330 | J | " |
| 0093 | 0393 | 0693 | 0993 | 3093 | 3393 | 3693 | 3993 | 330 | K | " |
| 0094 | 0394 | 0694 | 0994 | 3094 | 3394 | 3694 | 3994 | 390 | F | " |
| 0095 | 0395 | 0695 | 0995 | 3095 | 3395 | 3695 | 3995 | 390 | J | " |
| 0096 | 0396 | 0696 | 0996 | 3096 | 3396 | 3696 | 3996 | 390 | K | " |
| 0097 | 0397 | 0697 | 0997 | 3097 | 3397 | 3697 | 3997 | 470 | F | " |
| 0098 | 0398 | 0698 | 0998 | 3098 | 3398 | 3698 | 3998 | 470 | J | " |
| 0099 | 0399 | 0699 | 0999 | 3099 | 3399 | 3699 | 3999 | 470 | K | " |
| 0100 | 0400 | 0700 | 1000 | 3100 | 3400 | 3700 | 4000 | 560 | F | 100 |
| 0101 | 0401 | 0701 | 1001 | 3101 | 3401 | 3701 | 4001 | 560 | J | " |
| 0102 | 0402 | 0702 | 1002 | 3102 | 3402 | 3702 | 4002 | 560 | K | " |
| 0103 | 0403 | 0703 | 1003 | 3103 | 3403 | 3703 | 4003 | 680 | F | " |
| 0104 | 0404 | 0704 | 1004 | 3104 | 3404 | 3704 | 4004 | 680 | J | " |
| 0105 | 0405 | 0705 | 1005 | 3105 | 3405 | 3705 | 4005 | 680 | K | " |
| 0106 | 0406 | 0706 | 1006 | 3106 | 3406 | 3706 | 4006 | 820 | F | " |
| 0107 | 0407 | 0707 | 1007 | 3107 | 3407 | 3707 | 4007 | 820 | J | " |
| 0108 | 0408 | 0708 | 1008 | 3108 | 3408 | 3708 | 4008 | 820 | K | " |
| 0109 | 0409 | 0709 | 1009 | 3109 | 3409 | 3709 | 4009 | 1,000 | F | " |
| 0110 | 0410 | 0710 | 1010 | 3110 | 3410 | 3710 | 4010 | 1,000 | J | " |
| 0111 | 0411 | 0711 | 1011 | 3111 | 3411 | 3711 | 4011 | 1,000 | K | " |
| 0112 | 0412 | 0712 | 1012 | 3112 | 3412 | 3712 | 4012 | 1,200 | F | " |
| 0113 | 0413 | 0713 | 1013 | 3113 | 3413 | 3713 | 4013 | 1,200 | J | " |
| 0114 | 0414 | 0714 | 1014 | 3114 | 3414 | 3714 | 4014 | 1,200 | K | " |
| 0115 | 0415 | 0715 | 1015 | 3115 | 3415 | 3715 | 4015 | 1,500 | F | " |
| 0116 | 0416 | 0716 | 1016 | 3116 | 3416 | 3716 | 4016 | 1,500 | J | " |
| 0117 | 0417 | 0717 | 1017 | 3117 | 3417 | 3717 | 4017 | 1,500 | K | " |
| 0118 | 0418 | 0718 | 1018 | 3118 | 3418 | 3718 | 4018 | 1,800 | F | " |
| 0119 | 0419 | 0719 | 1019 | 3119 | 3419 | 3719 | 4019 | 1,800 | J | " |
| 0120 | 0420 | 0720 | 1020 | 3120 | 3420 | 3720 | 4020 | 1,800 | K | " |
| 0121 | 0421 | 0721 | 1021 | 3121 | 3421 | 3721 | 4021 | 2,200 | F | " |
| 0122 | 0422 | 0722 | 1022 | 3122 | 3422 | 3722 | 4022 | 2,200 | J | " |
| 0123 | 0423 | 0723 | 1023 | 3123 | 3423 | 3723 | 4023 | 2,200 | K | " |
| 0124 | 0424 | 0724 | 1024 | 3124 | 3424 | 3724 | 4024 | 2,700 | F | 50 |
| 0125 | 0425 | 0725 | 1025 | 3125 | 3425 | 3725 | 4025 | 2,700 | J | " |
| 0126 | 0426 | 0726 | 1026 | 3126 | 3426 | 3726 | 4026 | 2,700 | K | " |
| 0127 | 0427 | 0727 | 1027 | 3127 | 3427 | 3727 | 4027 | 3,300 | F | " |
| 0128 | 0428 | 0728 | 1028 | 3128 | 3428 | 3728 | 4028 | 3,300 | J | " |
| 0129 | 0429 | 0729 | 1029 | 3129 | 3429 | 3729 | 4029 | 3,300 | K | " |
| 0130 | 0430 | 0730 | 1030 | 3130 | 3430 | 3730 | 4030 | 3,900 | F | " |
| 0131 | 0431 | 0731 | 1031 | 3131 | 3431 | 3731 | 4031 | 3,900 | J | " |
| 0132 | 0432 | 0732 | 1032 | 3132 | 3432 | 3732 | 4032 | 3,900 | K | " |
| 0133 | 0433 | 0733 | 1033 | 3133 | 3433 | 3733 | 4033 | 4,700 | F | " |
| 0134 | 0434 | 0734 | 1034 | 3134 | 3434 | 3734 | 4034 | 4,700 | J | " |
| 0135 | 0435 | 0735 | 1035 | 3135 | 3435 | 3735 | 4035 | 4,700 | K | " |
| 0136 | 0436 | 0736 | 1036 | 3136 | 3436 | 3736 | 4036 | 5,600 | F | " |

See footnote at end of table.



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(C) TABLE III. Dash numbers and electrical characteristics - Continued.

| Dash number Failure rate level (%/1,000 hours) | | | | | | | | Capacitance (pF) | Capacitance tolerance | DC rated voltage volts) |
|---|---|---|---|---|---|---|---|---|---|---|
| Standard lead length 1/ | | | | Optional longer lead length 1/ | | | | | | |
| 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | | | |
| Style CKR22, voltage-temperature limits of 0 ±30 ppm/°C - Continued. | | | | | | | | | | |
| 0137 | 0437 | 0737 | 1037 | 3137 | 3437 | 3737 | 4037 | 5,600 | J | 50 |
| 0138 | 0438 | 0738 | 1038 | 3138 | 3438 | 3738 | 4038 | 5,600 | K | " |
| 0139 | 0439 | 0739 | 1039 | 3139 | 3439 | 3739 | 4039 | 6,800 | F | " |
| 0140 | 0440 | 0740 | 1040 | 3140 | 3440 | 3740 | 4040 | 6,800 | J | " |
| 0141 | 0441 | 0741 | 1041 | 3141 | 3441 | 3741 | 4041 | 6,800 | K | " |
| 0142 | 0442 | 0742 | 1042 | 3142 | 3442 | 3742 | 4042 | 8,200 | F | " |
| 0143 | 0443 | 0743 | 1043 | 3143 | 3443 | 3743 | 4043 | 8,200 | J | " |
| 0144 | 0444 | 0744 | 1044 | 3144 | 3444 | 3744 | 4044 | 8,200 | K | " |
| 0145 | 0445 | 0745 | 1045 | 3145 | 3445 | 3745 | 4045 | 10,000 | F | " |
| 0146 | 0446 | 0746 | 1046 | 3146 | 3446 | 3746 | 4046 | 10,000 | J | " |
| 0147 | 0447 | 0747 | 1047 | 3147 | 3447 | 3747 | 4047 | 10,000 | K | " |
| Style CKR22, voltage-temperature limits of ±15% and +15, -25% | | | | | | | | | | |
| 0148 | 0448 | 0748 | 1048 | 3148 | 3448 | 3748 | 4048 | 270 | K | 200 |
| 0149 | 0449 | 0749 | 1049 | 3149 | 3449 | 3749 | 4049 | 330 | K | " |
| 0150 | 0450 | 0750 | 1050 | 3150 | 3450 | 3750 | 4050 | 330 | M | " |
| 0151 | 0451 | 0751 | 1051 | 3151 | 3451 | 3751 | 4051 | 390 | K | " |
| 0152 | 0452 | 0752 | 1052 | 3152 | 3452 | 3752 | 4052 | 470 | K | " |
| 0153 | 0453 | 0753 | 1053 | 3153 | 3453 | 3753 | 4053 | 470 | M | " |
| 0154 | 0454 | 0754 | 1054 | 3154 | 3454 | 3754 | 4054 | 560 | K | " |
| 0155 | 0455 | 0755 | 1055 | 3155 | 3455 | 3755 | 4055 | 680 | K | " |
| 0156 | 0456 | 0756 | 1056 | 3156 | 3456 | 3756 | 4056 | 680 | M | " |
| 0157 | 0457 | 0757 | 1057 | 3157 | 3457 | 3757 | 4057 | 820 | K | " |
| 0158 | 0458 | 0758 | 1058 | 3158 | 3458 | 3758 | 4058 | 1,000 | K | 100 |
| 0159 | 0459 | 0759 | 1059 | 3159 | 3459 | 3759 | 4059 | 1,000 | M | " |
| 0160 | 0460 | 0760 | 1060 | 3160 | 3460 | 3760 | 4060 | 1,200 | K | " |
| 0161 | 0461 | 0761 | 1061 | 3161 | 3461 | 3761 | 4061 | 1,500 | K | " |
| 0162 | 0462 | 0762 | 1062 | 3162 | 3462 | 3762 | 4062 | 1,500 | M | " |
| 0163 | 0463 | 0763 | 1063 | 3163 | 3463 | 3763 | 4063 | 1,800 | K | " |
| 0164 | 0464 | 0764 | 1064 | 3164 | 3464 | 3764 | 4064 | 2,200 | K | " |
| 0165 | 0465 | 0765 | 1065 | 3165 | 3465 | 3765 | 4065 | 2,200 | M | " |
| 0166 | 0466 | 0766 | 1066 | 3166 | 3466 | 3766 | 4066 | 2,700 | K | " |
| 0167 | 0467 | 0767 | 1067 | 3167 | 3467 | 3767 | 4067 | 3,300 | K | " |
| 0168 | 0468 | 0768 | 1068 | 3168 | 3468 | 3768 | 4068 | 3,300 | M | " |
| 0169 | 0469 | 0769 | 1069 | 3169 | 3469 | 3769 | 4069 | 3,900 | K | " |
| 0170 | 0470 | 0770 | 1070 | 3170 | 3470 | 3770 | 4070 | 4,700 | K | " |
| 0171 | 0471 | 0771 | 1071 | 3171 | 3471 | 3771 | 4071 | 4,700 | M | " |
| 0172 | 0472 | 0772 | 1072 | 3172 | 3472 | 3772 | 4072 | 5,600 | K | " |
| 0173 | 0473 | 0773 | 1073 | 3173 | 3473 | 3773 | 4073 | 6,800 | K | " |
| 0174 | 0474 | 0774 | 1074 | 3174 | 3474 | 3774 | 4074 | 6,800 | M | " |
| 0175 | 0475 | 0775 | 1075 | 3175 | 3475 | 3775 | 4075 | 8,200 | K | " |
| 0176 | 0476 | 0776 | 1076 | 3176 | 3476 | 3776 | 4076 | 10,000 | K | " |
| 0177 | 0477 | 0777 | 1077 | 3177 | 3477 | 3777 | 4077 | 10,000 | M | " |
| 0178 | 0478 | 0778 | 1078 | 3178 | 3478 | 3778 | 4078 | 12,000 | K | 50 |
| 0179 | 0479 | 0779 | 1079 | 3179 | 3479 | 3779 | 4079 | 15,000 | K | " |
| 0180 | 0480 | 0780 | 1080 | 3180 | 3480 | 3780 | 4080 | 15,000 | M | " |
| 0181 | 0481 | 0781 | 1081 | 3181 | 3481 | 3781 | 4081 | 18,000 | K | " |
| 0182 | 0482 | 0782 | 1082 | 3182 | 3482 | 3782 | 4082 | 22,000 | K | " |
| 0183 | 0483 | 0783 | 1083 | 3183 | 3483 | 3783 | 4083 | 22,000 | M | " |
| 0184 | 0484 | 0784 | 1084 | 3184 | 3484 | 3784 | 4084 | 27,000 | K | " |
| 0185 | 0485 | 0785 | 1085 | 3185 | 3485 | 3785 | 4085 | 33,000 | K | " |
| 0186 | 0486 | 0786 | 1086 | 3186 | 3486 | 3786 | 4086 | 33,000 | M | " |
| 0187 | 0487 | 0787 | 1087 | 3187 | 3487 | 3787 | 4087 | 39,000 | K | " |
| 0188 | 0488 | 0788 | 1088 | 3188 | 3488 | 3788 | 4088 | 47,000 | K | " |
| 0189 | 0489 | 0789 | 1089 | 3189 | 3489 | 3789 | 4089 | 47,000 | M | " |
| 0190 | 0490 | 0790 | 1090 | 3190 | 3490 | 3790 | 4090 | 56,000 | K | " |
| 0191 | 0491 | 0791 | 1091 | 3191 | 3491 | 3791 | 4091 | 68,000 | K | " |

See footnote at end of table.



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(C) TABLE III. Dash numbers and electrical characteristics - Continued.

| Dash number Failure rate level (%/1,000 hours) | | | | | | | | Capacitance (pF) | Capacitance tolerance | DC rated voltage volts) |
|---|---|---|---|---|---|---|---|---|---|---|
| Standard lead length 1/ | | | | Optional longer lead length 1/ | | | | | | |
| 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | | | |
| Style CKR22, voltage-temperature limits of ±15% and +15, -25% - Continued. | | | | | | | | | | |
| 0192 | 0492 | 0792 | 1092 | 3192 | 3492 | 3792 | 4092 | 68,000 | M | 50 |
| 0193 | 0493 | 0793 | 1093 | 3193 | 3493 | 3793 | 4093 | 82,000 | K | " |
| 0194 | 0494 | 0794 | 1094 | 3194 | 3494 | 3794 | 4094 | 100,000 | K | " |
| 0195 | 0495 | 0795 | 1095 | 3195 | 3495 | 3795 | 4095 | 100,000 | M | " |
| Style CKR23, voltage-temperature limits of 0 ±30 ppm/°C | | | | | | | | | | |
| 0258 | 0558 | 0858 | 1158 | 3258 | 3558 | 3858 | 4158 | 560 | F | 200 |
| 0259 | 0559 | 0859 | 1159 | 3259 | 3559 | 3859 | 4159 | 560 | J | " |
| 0260 | 0560 | 0860 | 1160 | 3260 | 3560 | 3860 | 4160 | 560 | K | " |
| 0261 | 0561 | 0861 | 1161 | 3261 | 3561 | 3861 | 4161 | 680 | F | " |
| 0262 | 0562 | 0862 | 1162 | 3262 | 3562 | 3862 | 4162 | 680 | J | " |
| 0263 | 0563 | 0863 | 1163 | 3263 | 3563 | 3863 | 4163 | 680 | K | " |
| 0264 | 0564 | 0864 | 1164 | 3264 | 3564 | 3864 | 4164 | 820 | F | " |
| 0265 | 0565 | 0865 | 1165 | 3265 | 3565 | 3865 | 4165 | 820 | J | " |
| 0266 | 0566 | 0866 | 1166 | 3266 | 3566 | 3866 | 4166 | 820 | K | " |
| 0267 | 0567 | 0867 | 1167 | 3267 | 3567 | 3867 | 4167 | 1,000 | F | " |
| 0268 | 0568 | 0868 | 1168 | 3268 | 3568 | 3868 | 4168 | 1,000 | J | " |
| 0269 | 0569 | 0869 | 1169 | 3269 | 3569 | 3869 | 4169 | 1,000 | K | " |
| 0270 | 0570 | 0870 | 1170 | 3270 | 3570 | 3870 | 4170 | 1,200 | F | " |
| 0271 | 0571 | 0871 | 1171 | 3271 | 3571 | 3871 | 4171 | 1,200 | J | " |
| 0272 | 0572 | 0872 | 1172 | 3272 | 3572 | 3872 | 4172 | 1,200 | K | " |
| 0273 | 0573 | 0873 | 1173 | 3273 | 3573 | 3873 | 4173 | 2,700 | F | 100 |
| 0274 | 0574 | 0874 | 1174 | 3274 | 3574 | 3874 | 4174 | 2,700 | J | " |
| 0275 | 0575 | 0875 | 1175 | 3275 | 3575 | 3875 | 4175 | 2,700 | K | " |
| 0276 | 0576 | 0876 | 1176 | 3276 | 3576 | 3876 | 4176 | 3,300 | F | " |
| 0277 | 0577 | 0877 | 1177 | 3277 | 3577 | 3877 | 4177 | 3,300 | J | " |
| 0278 | 0578 | 0878 | 1178 | 3278 | 3578 | 3878 | 4178 | 3,300 | K | " |
| 0279 | 0579 | 0879 | 1179 | 3279 | 3579 | 3879 | 4179 | 4,700 | F | 50 |
| 0280 | 0580 | 0880 | 1180 | 3280 | 3580 | 3880 | 4180 | 4,700 | J | " |
| 0281 | 0581 | 0881 | 1181 | 3281 | 3581 | 3881 | 4181 | 4,700 | K | " |
| 0282 | 0582 | 0882 | 1182 | 3282 | 3582 | 3882 | 4182 | 5,600 | F | " |
| 0283 | 0583 | 0883 | 1183 | 3283 | 3583 | 3883 | 4183 | 5,600 | J | " |
| 0284 | 0584 | 0884 | 1184 | 3284 | 3584 | 3884 | 4184 | 5,600 | K | " |
| 0285 | 0585 | 0885 | 1185 | 3285 | 3585 | 3885 | 4185 | 6,800 | F | " |
| 0286 | 0586 | 0886 | 1186 | 3286 | 3586 | 3886 | 4186 | 6,800 | J | " |
| 0287 | 0587 | 0887 | 1187 | 3287 | 3587 | 3887 | 4187 | 6,800 | K | " |
| 0288 | 0588 | 0888 | 1188 | 3288 | 3588 | 3888 | 4188 | 8,200 | F | " |
| 0289 | 0589 | 0889 | 1189 | 3289 | 3589 | 3889 | 4189 | 8,200 | J | " |
| 0290 | 0590 | 0890 | 1190 | 3290 | 3590 | 3890 | 4190 | 8,200 | K | " |
| 0291 | 0591 | 0891 | 1191 | 3291 | 3591 | 3891 | 4191 | 10,000 | F | " |
| 0292 | 0592 | 0892 | 1192 | 3292 | 3592 | 3892 | 4192 | 10,000 | J | " |
| 0293 | 0593 | 0893 | 1193 | 3293 | 3593 | 3893 | 4193 | 10,000 | K | " |
| Style CKR23, voltage-temperature limits of +15% and +15, -25% - Continued. | | | | | | | | | | |
| 0196 | 0496 | 0796 | 1096 | 3196 | 3496 | 3796 | 4096 | 1,000 | K | 200 |
| 0197 | 0497 | 0797 | 1097 | 3197 | 3497 | 3797 | 4097 | 1,000 | M | " |
| 0198 | 0498 | 0798 | 1098 | 3198 | 3498 | 3798 | 4098 | 1,200 | K | " |
| 0199 | 0499 | 0799 | 1099 | 3199 | 3499 | 3799 | 4099 | 1,500 | K | " |
| 0200 | 0500 | 0800 | 1100 | 3200 | 3500 | 3800 | 4100 | 1,500 | M | " |
| 0201 | 0501 | 0801 | 1101 | 3201 | 3501 | 3801 | 4101 | 1,800 | K | " |
| 0202 | 0502 | 0802 | 1102 | 3202 | 3502 | 3802 | 4102 | 2,200 | K | " |
| 0203 | 0503 | 0803 | 1103 | 3203 | 3503 | 3803 | 4103 | 2,200 | M | " |
| 0204 | 0504 | 0804 | 1104 | 3204 | 3504 | 3804 | 4104 | 2,700 | K | " |
| 0205 | 0505 | 0805 | 1105 | 3205 | 3505 | 3805 | 4105 | 3,300 | K | " |
| 0206 | 0506 | 0806 | 1106 | 3206 | 3506 | 3806 | 4106 | 3,300 | M | " |

See footnote at end of table.



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© TABLE III. Dash numbers and electrical characteristics - Continued.

| Dash number Failure rate level (%/1,000 hours) | | | | | | | | Capacitance (pF) | Capacitance tolerance | DC rated voltage volts) |
|---|---|---|---|---|---|---|---|---|---|---|
| Standard lead length 1/ | | | | Optional longer lead length 1/ | | | | | | |
| 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | 1.0(M) | 0.1(P) | 0.01(R) | 0.001(S) | | | |
| Style CKR23, voltage-temperature limits of +15% and +15, -25% - Continued. | | | | | | | | | | |
| 0207 | 0507 | 0807 | 1107 | 3207 | 3507 | 3807 | 4107 | 3,900 | K | 200 |
| 0208 | 0508 | 0808 | 1108 | 3208 | 3508 | 3808 | 4108 | 4,700 | K | " |
| 0209 | 0509 | 0809 | 1109 | 3209 | 3509 | 3809 | 4109 | 4,700 | M | " |
| 0210 | 0510 | 0810 | 1110 | 3210 | 3510 | 3810 | 4110 | 5,600 | K | " |
| 0211 | 0511 | 0811 | 1111 | 3211 | 3511 | 3811 | 4111 | 6,800 | K | " |
| 0212 | 0512 | 0812 | 1112 | 3212 | 3512 | 3812 | 4112 | 6,800 | M | " |
| 0213 | 0513 | 0813 | 1113 | 3213 | 3513 | 3813 | 4113 | 8,200 | K | " |
| 0214 | 0514 | 0814 | 1114 | 3214 | 3514 | 3814 | 4114 | 10,000 | K | " |
| 0215 | 0515 | 0815 | 1115 | 3215 | 3515 | 3815 | 4115 | 10,000 | M | " |
| 0216 | 0516 | 0816 | 1116 | 3216 | 3516 | 3816 | 4116 | 12,000 | K | 100 |
| 0217 | 0517 | 0817 | 1117 | 3217 | 3517 | 3817 | 4117 | 15,000 | K | " |
| 0218 | 0518 | 0818 | 1118 | 3218 | 3518 | 3818 | 4118 | 15,000 | M | " |
| 0219 | 0519 | 0819 | 1119 | 3219 | 3519 | 3819 | 4119 | 18,000 | K | " |
| 0220 | 0520 | 0820 | 1120 | 3220 | 3520 | 3820 | 4120 | 22,000 | K | " |
| 0221 | 0521 | 0821 | 1121 | 3221 | 3521 | 3821 | 4121 | 22,000 | M | " |
| 0222 | 0522 | 0822 | 1122 | 3222 | 3522 | 3822 | 4122 | 27,000 | K | " |
| 0223 | 0523 | 0823 | 1123 | 3223 | 3523 | 3823 | 4123 | 33,000 | K | " |
| 0224 | 0524 | 0824 | 1124 | 3224 | 3524 | 3824 | 4124 | 33,000 | M | " |
| 0225 | 0525 | 0825 | 1125 | 3225 | 3525 | 3825 | 4125 | 39,000 | K | " |
| 0226 | 0526 | 0826 | 1126 | 3226 | 3526 | 3826 | 4126 | 47,000 | K | " |
| 0227 | 0527 | 0827 | 1127 | 3227 | 3527 | 3827 | 4127 | 47,000 | M | " |
| 0228 | 0528 | 0828 | 1128 | 3228 | 3528 | 3828 | 4128 | 56,000 | K | " |
| 0229 | 0529 | 0829 | 1129 | 3229 | 3529 | 3829 | 4129 | 68,000 | K | " |
| 0230 | 0530 | 0830 | 1130 | 3230 | 3530 | 3830 | 4130 | 68,000 | M | " |
| 0231 | 0531 | 0831 | 1131 | 3231 | 3531 | 3831 | 4131 | 82,000 | K | " |
| 0232 | 0532 | 0832 | 1132 | 3232 | 3532 | 3832 | 4132 | 100,000 | K | " |
| 0233 | 0533 | 0833 | 1133 | 3233 | 3533 | 3833 | 4133 | 100,000 | M | " |
| 0234 | 0534 | 0834 | 1134 | 3234 | 3534 | 3834 | 4134 | 120,000 | K | 50 |
| 0235 | 0535 | 0835 | 1135 | 3235 | 3535 | 3835 | 4135 | 150,000 | K | " |
| 0236 | 0536 | 0836 | 1136 | 3236 | 3536 | 3836 | 4136 | 150,000 | M | " |
| 0237 | 0537 | 0837 | 1137 | 3237 | 3537 | 3837 | 4137 | 180,000 | K | " |
| 0238 | 0538 | 0838 | 1138 | 3238 | 3538 | 3838 | 4138 | 220,000 | K | " |
| 0239 | 0539 | 0839 | 1139 | 3239 | 3539 | 3839 | 4139 | 220,000 | M | " |
| Style CKR24, voltage-temperature limits of ±15% and +15, -40% | | | | | | | | | | |
| 0240 | 0540 | 0840 | 1140 | 3240 | 3540 | 3840 | 4140 | 120,000 | K | 100 |
| 0241 | 0541 | 0841 | 1141 | 3241 | 3541 | 3841 | 4141 | 150,000 | K | 100 |
| 0242 | 0542 | 0842 | 1142 | 3242 | 3542 | 3842 | 4142 | 150,000 | M | 100 |
| 0243 | 0543 | 0843 | 1143 | 3243 | 3543 | 3843 | 4143 | 180,000 | K | 50 |
| 0244 | 0544 | 0844 | 1144 | 3244 | 3544 | 3844 | 4144 | 220,000 | K | " |
| 0245 | 0545 | 0845 | 1145 | 3245 | 3545 | 3845 | 4145 | 220,000 | M | " |
| 0246 | 0546 | 0846 | 1146 | 3246 | 3546 | 3846 | 4146 | 270,000 | K | " |
| 0247 | 0547 | 0847 | 1147 | 3247 | 3547 | 3847 | 4147 | 330,000 | K | " |
| 0248 | 0548 | 0848 | 1148 | 3248 | 3548 | 3848 | 4148 | 330,000 | M | " |
| 0249 | 0549 | 0849 | 1149 | 3249 | 3549 | 3849 | 4149 | 390,000 | K | " |
| 0250 | 0550 | 0850 | 1150 | 3250 | 3550 | 3850 | 4150 | 470,000 | K | " |
| 0251 | 0551 | 0851 | 1151 | 3251 | 3551 | 3851 | 4151 | 470,000 | M | " |
| 0252 | 0552 | 0852 | 1152 | 3252 | 3552 | 3852 | 4152 | 560,000 | K | " |
| 0253 | 0553 | 0853 | 1153 | 3253 | 3553 | 3853 | 4153 | 680,000 | K | " |
| 0254 | 0554 | 0854 | 1154 | 3254 | 3554 | 3854 | 4154 | 680,000 | M | " |
| 0255 | 0555 | 0855 | 1155 | 3255 | 3555 | 3855 | 4155 | 820,000 | K | " |
| 0256 | 0556 | 0856 | 1156 | 3256 | 3556 | 3856 | 4156 | 1,000,000 | K | " |
| 0257 | 0557 | 0857 | 1157 | 3257 | 3557 | 3857 | 4157 | 1,000,000 | M | " |

1/ See figure 1 for differences in lead length. The optional longer lead length is, with acquiring activity approval, substitutable for the same part with standard lead length.



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-1840)

Review activities:
Army - AR, MI
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19



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